
Ecology & Environment MCQs

1. **Ecology** is the biological study of living organisms and their interaction with each other and their environment.
Answer: Ecology
2. **Lotic freshwater zones** are flowing water systems.
Answer: Flowing water
3. **Mining waste is generated in** how many ways?
Answer: 3 ways
4. **Rise in temperature over the past 100 years** is due to?
Answer: Global warming
5. **More than _____ of water is groundwater.**
Answer: 30%
6. **Lotic freshwater zones are**
Answer: Flowing water zones (e.g., streams, rivers)
7. **Which ecosystem is known as the land of midnight sun?**
Answer: Tundra ecosystem
8. **Climatic zones of Pakistan:**
Answer: 4
9. **The snow leopard is still occasionally called:**
Answer: Ounce
10. **Animals living at high altitudes include:**
Answer: All (Snow Leopard, Snow Pigeon, Snow Partridge)
11. **The average temperature of Earth's surface is increasing due to:**
Answer: Global warming
12. **Types of pollution:**
Answer: 10
13. **Mining waste is generated by:**
Answer: 3 ways
14. **Water present below Earth's surface is called:**
Answer: Groundwater
15. **Water lily is a:**
Answer: Floating plant
16. **In aquatic ecosystems, plants present are:**
Answer: Both a and b (Floating and submerged)
17. **Laws of thermodynamics describe relationship between:**
Answer: Thermal energy and other forms of energy
18. **Which one is not a type of symbiosis?**
Answer: Lion/Deer
19. **Runoff from urban and suburban areas is a major origin of:**
Answer: Non-point source pollution

20. **Particles blocked by skin and paper are:**
Answer: Alpha particles
21. **As temperature of water increases, dissolved oxygen:**
Answer: Decreases
22. **Industrial noise pollution damages hearing ability by:**
Answer: Around 20%
23. **Choose the abiotic components:**
Answer: Climatic, edaphic, social
24. **Markhor is present in:**
Answer: Northern mountains of Pakistan
25. **Ecosystem with highest biodiversity:**
Answer: Marine ecosystem
26. **Light clutter refers to:**
Answer: Excessive groupings of lights
27. **Not present in Himalayan forests:**
Answer: Stepped eagle
28. **Birds with colorful plumage:**
Answer: Peacock, Parrot (Example; actual options not given)
29. **Typified areas between Jhang and Shorkot Road:**
Answer: Indus Plain
30. **Light pollution in Hong Kong (March 2013) declared:**
Answer: Worst on the planet
31. **Municipal solid waste (per capita in grams):**
Answer: 240 grams
32. **Effect of climate change on political issues:**
Answer: Conflict over resources, migration (Examples)
33. **Method to reduce acid rain effect on soil:**
Answer: Add lime (liming)
34. **Natural cause of smog:**
Answer: Volcanic activity
35. **Significance of nitrogen fixation by cyanobacteria in rice paddies:**
Answer: Enriches soil with nitrogen
36. **Parasitic weed:**
Answer: Dodder (Cuscuta)
37. **Interspecific interaction with long-term association:**
Answer: Symbiosis
38. **Layer closest to Earth's surface:**
Answer: Troposphere
39. **Light trespass is:**
Answer: Unwanted light entering areas like bedrooms at night
40. **Height of K2:**
Answer: 8611 meters
41. **Sign of bravery and competition:**
Answer: Markhor
42. **Percentage of Earth's surface that is seawater:**
Answer: 70%

43. Layer of water mixed by wind & warmed by sun (spring to fall):

Answer: Epilimnion

44. Chinese zodiac includes how many animals?

Answer: 12

45. Climatic zones of Pakistan:

Answer: 4

46. Types of pollution:

Answer: 10

47. Animal that hibernates in winter:

Answer: Frog

48. Tropical grasslands are called:

Answer: Savannas

49. Nitrogen fixation bacteria:

Answer: Nitrobacter

50. Growth curves can be:

Answer: J and S (sigmoid) shaped

51. 85% of phosphorus is found in:

Answer: Bones and teeth

52. First trophic level are:

Answer: Autotrophs

53. Types of ecosystems:

Answer: Natural and artificial

54. Limiting factor for humans:

Answer: Temperature (37°C)

55. Example of floating plant phytoplankton:

Answer: Spirogyra, Volvox, Diatoms

56. Sign of parasitism:

Answer: (+/-)

57. Ecology describes richness of:

Answer: Biosphere

58. Largest desert:

Answer: Sahara

59. Tropical rainforest feature:

Answer: High rainfall

60. Percentage of ocean water:

Answer: 97.3%

61. Amensalism describes:

Answer: One species negatively affects another

62. Natality refers to:

Answer: Population birth rate or ability to increase

63. Ecology is the study of:

Answer: Organisms and their environment

64. Ecosystem consists of interactions between:

Answer: Biotic and Abiotic components

65. Factor of ecosystem:

Answer: Both biotic and abiotic

Let me know if you'd like this in a PDF or formatted worksheet!

Questions:

Short question

1. Population growth curve
2. What is decomposers
3. Climate Of Coniferous forest
 1. Steps involved in Carbon Cycle
 2. Amensalism
 3. Prairies, Savanna & Stepes
 4. How miming waste is generated
 5. What is the surface ground pollution write it's cause.
- 1.scope of ecology?
- 2.what is IDA?
- 3.Define pollution? Write three types?
- 4.Markhore distribution in pak?

Here's a detailed, easy-to-understand and organized response to all your questions, categorized under proper headings:

1. Population Growth Curve

A **population growth curve** is a graphical representation showing how the population size of an organism changes over time. It typically follows two patterns:

- **Lag Phase:** Slow growth at the beginning.
 - **Exponential Phase:** Rapid increase in population.
 - **Stationary Phase:** Population growth slows as resources become limited.
 - **Decline Phase** (if applicable): Population may decrease due to scarcity of resources or disease.
-

2. What are Decomposers?

Decomposers are organisms that break down dead and decaying plants and animals into simpler substances.

Examples: **Bacteria**, **fungi**, and some insects.

They play a vital role in **nutrient recycling** in ecosystems.

3. Climate of Coniferous Forest

Coniferous forests (also called **taiga** or **boreal forests**) have:

- **Cold winters** and **mild summers**
 - **Low precipitation** (mostly snow)
 - **Short growing season**
 - **Soil** is acidic and nutrient-poor due to slow decomposition.
-

4. Steps Involved in Carbon Cycle

1. **Photosynthesis** – Plants absorb CO₂ and convert it into glucose.
 2. **Respiration** – Animals and plants release CO₂ back into the air.
 3. **Decomposition** – Decomposers break down dead matter, releasing CO₂.
 4. **Combustion** – Burning fossil fuels releases stored carbon.
 5. **Ocean Absorption** – Oceans absorb CO₂ from the atmosphere.
-

5. Amensalism

Amensalism is a type of interaction where one organism is harmed while the other is unaffected. Example: Penicillium fungus produces antibiotics that kill bacteria, but the fungus itself is not affected.

6. Prairies, Savanna & Steppes

- **Prairies:** Temperate grasslands found in North America; rich in grasses and very fertile soil.
 - **Savannas:** Tropical grasslands with scattered trees; found in Africa; have dry and wet seasons.
 - **Steppes:** Dry grasslands in Europe and Asia; have cold winters and hot summers; fewer trees.
-

7. How Mining Waste is Generated

Mining waste is generated during:

- **Extraction** of minerals (removal of topsoil, rocks)
 - **Processing** (chemical treatment)
 - **Storage** of leftover materials like tailings and slag.
This waste may contain **heavy metals, toxins**, and non-biodegradable materials.
-

8. What is Surface Ground Pollution? Causes?

Surface ground pollution refers to contamination of the soil and land surface.

Causes:

- Dumping of industrial or household waste
 - Agricultural chemicals (pesticides, fertilizers)
 - Mining and construction activities
 - Oil spills and e-waste
-

9. Scope of Ecology

Ecology studies the relationships between living organisms and their environment.

Scope includes:

- Ecosystem structure and function
 - Biodiversity and conservation
 - Population and community dynamics
 - Environmental pollution and sustainable practices
-

10. What is IDA?

The International Dark-Sky Association, formally known today as DarkSky International, was founded in 1988 by David Crawford, a professional astronomer, and Tim Hunter, a physician and amateur .

Its mission is to preserve and protect the nighttime environment and our heritage of dark skies through quality outdoor lighting practices, public policy engagement, education, and outreach

11. Define Pollution and Its Types

Pollution is the introduction of harmful substances or energy into the environment that causes adverse effects.

Three Types:

1. **Air Pollution** (e.g., from factories, vehicles)
 2. **Water Pollution** (e.g., dumping waste in rivers)
 3. **Soil Pollution** (e.g., pesticides, landfills)
-

12. Markhor Distribution in Pakistan

The **Markhor**, Pakistan's national animal, is found in:

- **Northern and western mountains** of Pakistan
 - Areas include **Chitral, Gilgit-Baltistan, Kohistan**, and **Torghar Hills** in Balochistan. They live in rugged, rocky terrains at high altitudes.
-

Let me know if you need diagrams or assignment formatting for any of these topics!

LQLong question

. Threats to Snow Leopard

Snow leopards, an endangered species, face several threats:

- **Habitat loss** due to infrastructure development (roads, mining).
 - **Poaching** for their fur, bones, and other body parts.
 - **Human-wildlife conflict** when snow leopards attack livestock.
 - **Climate change**, which reduces the availability of prey and suitable habitat.
 - **Lack of genetic diversity** due to isolated populations.
-

2. Define Glare & Its Three Types

Glare is excessive or uncontrolled brightness that causes discomfort or reduces visibility.

Three types of glare:

1. **Disability Glare** – Reduces the ability to see (e.g., oncoming car headlights at night).
 2. **Discomfort Glare** – Causes irritation but doesn't affect vision much.
 3. **Reflected Glare** – Caused by light reflecting off shiny surfaces (e.g., sunlight on water or screens).
-

3. How Community Ecology Helps Understand Species Interactions

Community ecology studies how different species interact within a shared environment.

It helps in understanding:

- **Predator-prey relationships**
- **Competition** for resources
- **Symbiosis** (mutualism, commensalism, parasitism)
- **Food chains and webs**
- **Species diversity** and ecosystem stability

This knowledge is useful in **conservation, ecosystem management, and biodiversity protection.**

4. Types of Acid Rain

Acid rain occurs when sulfur dioxide (SO₂) and nitrogen oxides (NO_x) mix with rainwater.

Types:

1. **Wet Acid Rain** – Rain, snow, or fog that contains acidic pollutants.
2. **Dry Acid Rain** – Acidic gases or particles that settle on surfaces without precipitation.

Both forms can damage **plants, soil, water bodies, and infrastructure**.

5. Why is Phosphorus Important?

Phosphorus is an **essential nutrient** for all living organisms.

- It's a key part of **DNA and RNA**, which carry genetic information.
- It forms **ATP (adenosine triphosphate)** – the energy currency of cells.
- Supports **root development, flowering, and seed formation** in plants.
- Helps in **bone and teeth** formation in animals and humans.

A deficiency in phosphorus can limit **growth and development** in both plants and animals.

Let me know if you'd like this in assignment format (with headings, bullet points, or MS Word style)!

2. What is recycling? Write benefits of recycling.

3. Resource Partitioning

4. Ecosphere

5. Solution of land pollution .

5. Which is not commensalism *bee and flower*

8. Underground water pollution and its causes.

Here are well-explained answers to your questions in a clear and student-friendly way:

1. What is Recycling? Write Benefits of Recycling.

Recycling is the process of collecting, processing, and reusing waste materials to create new products instead of throwing them away.

Benefits of Recycling:

- **Reduces pollution** (air, water, and land)
 - **Conserves natural resources** (like timber, water, and minerals)
 - **Saves energy** (less energy needed than making products from raw materials)
 - **Reduces landfill waste**
 - **Creates jobs** in recycling industries
 - **Fights climate change** by reducing greenhouse gas emissions
-

2. Resource Partitioning

Resource partitioning is the process by which species avoid competition by using the environment differently.

Example: Different bird species may live in the same tree but feed at different heights or times of the day.

It helps in:

- **Reducing direct competition**
 - **Allowing coexistence of multiple species**
 - **Maintaining biodiversity**
-

3. Ecosphere

The **Ecosphere** refers to the global ecosystem – the part of the Earth where life exists, including:

- **Atmosphere** (air)
- **Hydrosphere** (water)
- **Lithosphere** (land)
- **Biosphere** (living organisms)

It represents the **interconnected system of all ecosystems** on Earth.

4. Solutions of Land Pollution

To reduce **land pollution**, we can:

- **Recycle and reuse** materials
 - **Properly dispose of waste** (especially hazardous waste)
 - **Reduce plastic usage**
 - **Use organic fertilizers** and avoid chemical pesticides
 - **Plant trees** and restore degraded land
 - **Promote sustainable farming** and waste management
-

5. Which is Not Commensalism: Bee and Flower?

Yes, the **bee and flower relationship is NOT commensalism** — it is an example of **mutualism**.

Why?

- The bee gets nectar (food).
- The flower gets pollinated.

Mutualism benefits both organisms, while **commensalism** benefits one without affecting the other.

6. Underground Water Pollution and Its Causes

Underground water pollution occurs when harmful substances seep into aquifers and contaminate groundwater.

Causes:

- **Leaking underground storage tanks** (e.g., oil or chemical tanks)
- **Excessive use of fertilizers and pesticides** in agriculture
- **Industrial waste dumping**

- **Sewage leakage and septic tank failure**
- **Landfills and waste disposal sites**

It affects drinking water quality and can lead to serious health issues.

1. Distribution of marine ecosystem.
2. Radioactive contamination.
3. Point Pollutants and it's 3 examples.
4. Causes of noise Pollution.

Here are clear and organized answers to your questions with headings for easy understanding:

1. Distribution of Marine Ecosystem

The **marine ecosystem** is the largest aquatic ecosystem on Earth and is distributed across various zones based on depth, sunlight, and distance from shore. It includes:

1. **Coastal Zone**
 - Shallow areas near shores, including estuaries, mangroves, and coral reefs.
 - High biodiversity due to sunlight and nutrients.
 2. **Pelagic Zone**
 - Open ocean away from the coast.
 - Divided into photic (sunlit) and aphotic (dark) zones.
 3. **Benthic Zone**
 - Ocean floor, including deep-sea trenches and hydrothermal vents.
 - Low light, cold temperatures, and high pressure.
 4. **Intertidal Zone**
 - Area between high and low tide lines.
 - Organisms must adapt to changing water levels and wave action.
-

2. Radioactive Contamination

Radioactive contamination is the presence of radioactive substances in the environment, which can harm living organisms and ecosystems.

Sources include:

- **Nuclear power plant accidents** (e.g., Chernobyl, Fukushima)
- **Improper disposal** of radioactive waste
- **Nuclear weapons testing**
- **Medical or industrial leaks**

Effects:

- Genetic mutations
 - Cancer and health disorders
 - Long-term damage to soil, water, and food chains
-

3. Point Pollutants and 3 Examples

Point pollutants are pollutants that come from a single, identifiable source.

Examples:

1. **Factory discharge pipe** releasing chemicals into a river
2. **Oil leak from a pipeline**
3. **Sewage outlet** draining into a water body

Point pollution is easier to monitor and control compared to non-point pollution.

4. Causes of Noise Pollution

Noise pollution is unwanted or harmful sound that disrupts normal activities and health.

Causes include:

- **Traffic and transport** (cars, trains, airplanes)
- **Industrial machinery** in factories
- **Construction activities** (drilling, hammering)
- **Loudspeakers, music, and public events**
- **Household appliances** (generators, mixers, TVs at high volume)

Effects: Hearing loss, stress, sleep disturbances, and behavioral issues.

Let me know if you'd like a combined PDF or assignment version of all these answers!

Here are the detailed and easy-to-understand answers to your questions with proper headings:

1. Municipal Solid Waste (MSW)

Municipal Solid Waste refers to the everyday garbage or trash generated by households, commercial establishments, institutions, and municipal services.

Examples include:

- Food waste
- Paper and packaging
- Plastic, glass, metal
- Yard waste (leaves, branches)
- Electronics and household items

Problems caused by MSW:

- Land pollution
- Foul odors and pests
- Water contamination (leachate)
- Greenhouse gas emissions from landfills (like methane)

Solutions:

- Waste segregation
 - Recycling and composting
 - Reducing plastic use
 - Energy recovery from waste (waste-to-energy plants)
-

2. Geographical Distribution of Markhor in Pakistan

Markhor (*Capra falconeri*), the national animal of Pakistan, is found in mountainous regions of the country.

Distribution Areas:

- **Northern and Western Pakistan**
 - **Gilgit-Baltistan**
 - **Chitral** (Khyber Pakhtunkhwa)
 - **Swat and Kohistan**
 - **Torghar Hills** (northern Balochistan)
 - **Hingol National Park** (some isolated populations)

Habitat:

- Steep rocky cliffs, dry scrub forests, and alpine terrain at elevations from 600 to 3,600 meters.
-

3. Two Examples of Amensalism

Amensalism is a relationship where one organism is harmed and the other is unaffected.

Examples:

1. **Penicillium fungus and bacteria**
 - Penicillium releases antibiotics that kill bacteria, but the fungus is unaffected.
 2. **Black walnut tree and nearby plants**
 - It releases a chemical called *juglone* that inhibits the growth of other plants around it.
-

4. Health Effects Due to Climate Change

Climate change negatively impacts human health in various ways:

- **Heat stress and heatstroke** from rising temperatures
- **Increased respiratory diseases** (e.g., asthma) due to poor air quality
- **Spread of infectious diseases** like malaria and dengue due to expanding habitats of mosquitoes
- **Malnutrition** caused by crop failure or reduced food security
- **Mental health issues** due to disasters (e.g., floods, wildfires)

- **Waterborne diseases** from contaminated water sources (due to floods or droughts)
-

Here are detailed and well-organized answers to your questions, ideal for assignments or exam preparation:

1. Threats to Snow Leopards

Snow leopards (*Panthera uncia*) are endangered and face several serious threats:

a. Habitat Loss

- Due to infrastructure development (roads, dams, mining)
- Human encroachment into mountainous areas

b. Poaching and Illegal Trade

- Hunted for their fur, bones, and other body parts used in traditional medicine.

c. Human-Wildlife Conflict

- When snow leopards attack livestock, herders often kill them in retaliation.

d. Climate Change

- Rising temperatures affect the snow line and reduce the snow leopard's natural habitat.
- Prey species like ibex and blue sheep also decline.

e. Low Genetic Diversity

- Small, fragmented populations reduce breeding success and adaptability.
-

2. Community Structure (In Detail)

Community structure refers to the composition, organization, and relationships among species living together in a habitat.

Key Elements of Community Structure:

1. **Species Composition**
 - The different species present in a community.
2. **Species Diversity**
 - Includes:
 - **Species richness:** Number of species
 - **Species evenness:** Relative abundance of each species
3. **Trophic Structure**
 - The feeding relationships:
 - Producers → Herbivores → Carnivores → Decomposers
4. **Dominant Species**
 - Species that are most abundant or have the most biomass.
5. **Keystone Species**
 - Species that have a disproportionately large effect on their community (e.g., predators or pollinators).
6. **Ecological Niches**
 - Each species' role in the ecosystem (what it eats, where it lives, etc.)
7. **Stratification**
 - Vertical layering in ecosystems (e.g., forest canopy, understory, ground level)
8. **Succession**
 - Natural changes in community structure over time (primary and secondary succession)

3. Effects of Radiation

Radiation can be natural (e.g., UV rays, radon gas) or artificial (e.g., nuclear accidents, medical radiation).

Effects on Humans and Organisms:

- **Cell damage and mutations**
- **Cancer development** (e.g., skin, thyroid, lung)
- **Radiation sickness** (nausea, fatigue, organ failure)
- **Birth defects and infertility**
- **Death of plants and animals** at high exposure levels

Environmental Effects:

- Long-term soil and water contamination
 - Death or mutation in wildlife populations
 - Reduced agricultural productivity
-

4. Effects of Climate Change

Climate change causes global and regional environmental shifts that impact ecosystems, biodiversity, and human health.

Key Effects:

1. **Global Warming**
 - Rising temperatures causing melting glaciers and polar ice
 2. **Extreme Weather Events**
 - More frequent floods, droughts, wildfires, and hurricanes
 3. **Sea Level Rise**
 - Due to melting ice caps and thermal expansion of oceans
 - Threatens coastal cities and islands
 4. **Loss of Biodiversity**
 - Extinction of species unable to adapt to changing climates
 5. **Ocean Acidification**
 - CO₂ dissolves in oceans, harming coral reefs and marine life
 6. **Impact on Agriculture**
 - Crop failure, food insecurity, and reduced yields
 7. **Human Health Risks**
 - Spread of vector-borne diseases (e.g., malaria, dengue)
 - Respiratory problems from poor air quality
 - Water scarcity and malnutrition
-

Let me know if you'd like this formatted into an assignment-style Word document or PDF!